

STAT 341: Tutorial Test 2 - F19

First name

Last name

Signature

ID Number

Email

Time allowed: 50 minutes

Special instructions

- All numeric answers should be written either as a number to three decimal places or as a fraction reduced to its lowest terms.
- If working is not specifically requested, a correct answer will receive full marks. However, an incorrect answer could still receive credit if working is provided. An incorrect answer with no working will receive zero marks.

1. [**1 mark**] Provide one reason as to why probabilistic sampling is superior to non-probabilistic sampling.
2. [**2 marks**] Consider taking a simple random sample of size n *with* replacement from a population of N units. Derive the joint probability of inclusion.
3. [**3 marks**] Consider the population attribute $\alpha(\mathcal{P})$. Based on a random sample $\mathcal{S}$, the population attribute is estimated by the sample attribute $\alpha(S) = \tilde{\alpha}(\mathcal{P})$. Show that

4. (a) [**3 marks**] Show that the HT estimator $\sum_{u \in \mathcal{S}} y_u / \pi_u$ is unbiased for the population total. Clearly define any notation used.

- (b) [**2 marks**] Based on the sample $\mathcal{S}$, write down an estimate of the variance of the Horvitz-Thompson estimator $a_{HT}(\mathcal{S})$,

$$Var(a_{HT}(\mathcal{S})) = \sum_{u \in \mathcal{P}} \sum_{v \in \mathcal{P}} (\pi_{uv} - \pi_u \pi_v) \frac{y_u}{\pi_u} \frac{y_v}{\pi_v}.$$

5. A sample ($n = 6$) was randomly selected from a population ($N = 124$). The inclusion probabilities and the observed y values from the sample are

y_i	-2	-1	0	1	2	3
π_i	4/50	4/50	4/50	4/50	1/73	1

- (a) [**2 marks**] Give the HT estimate for the proportion of units less than or equal to -2 .

- (b) [**2 marks**] Give the HT estimate for the population average.

- (c) [**2 marks**] Give the HT estimate of the population median.

6. (a) [4 marks] Describe in point form a test of significance using a discrepancy measure $D(\mathcal{P}_1, \mathcal{P}_2)$

(b) [1 mark] What is the observed significance level?

7. [6 marks (1 mark each)] Determine whether the following statements are true or false.

- (a) When sampling with replacement the sample average is a Horvitz-Thompson estimate.
 - i. True
 - ii. False

- (b) The Horvitz-Thompson estimator of the population mean is a scale equivariant estimator.
 - i. True
 - ii. False

- (c) The difference between the attribute of the study population and that of the sample is called the study error.
 - i. True
 - ii. False

- (d) The observed significance level measures evidence in favour of the hypothesis.
 - i. True
 - ii. False

- (e) The relative error of an estimator is a number between 0 and 1.
 - i. True
 - ii. False

- (f) We can quantify the sampling error by calculating all possible samples.
 - i. True
 - ii. False